

V3.9 BOM 核对清单（按位号顺序）

共 178 个位号，其中 157 个需要贴片。贴完一个在左侧方框打勾。坐标以 PCB 左上角为原点，单位 mm。

板上丝印印错了 —— 别信板上的字，认坐标

有极性/方向 —— 贴反不工作

不贴

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	C1	10uF	C_0805_2012Metric	78.6, 93.3	A 电源	
	C2	1uF	C_0603_1608Metric	62.0, 98.3	A 电源	
	C3	1uF	C_0603_1608Metric	66.5, 98.3	A 电源	
	C4	1uF	C_0603_1608Metric	71.0, 98.3	A 电源	
	C5	1uF	C_0402_1005Metric	109.6, 79.2	A 电源	
	C6	1uF	C_0603_1608Metric	78.6, 67.0	A 电源	板上印着 C18
	C10	100nF	C_0603_1608Metric	66.4, 84.5	C 传感器 + GNSS	
	C11	100nF	C_0603_1608Metric	70.6, 84.5	C 传感器 + GNSS	
	C12	100nF	C_0603_1608Metric	74.8, 84.5	C 传感器 + GNSS	
	C13	100nF	C_0603_1608Metric	79.1, 84.5	C 传感器 + GNSS	
	C14	100nF	C_0603_1608Metric	61.8, 87.2	C 传感器 + GNSS	
	C15	4.7uF	C_0603_1608Metric	66.1, 87.2	C 传感器 + GNSS	
	C16	100nF	C_0603_1608Metric	70.4, 87.2	C 传感器 + GNSS	
	C17	100nF	C_0603_1608Metric	78.5, 61.4	C 传感器 + GNSS	
	C18	10uF	C_0805_2012Metric	78.9, 64.6	C 传感器 + GNSS	板上印着 C6
	C19	15pF C0G	C_0603_1608Metric	104.6, 87.7	B MCU + Flash	
	C20	15pF C0G	C_0603_1608Metric	107.6, 87.7	B MCU + Flash	
	C21	100pF	C_0402_1005Metric	101.0, 67.5	E 1090 接收链	
	C22	100nF	C_0603_1608Metric	95.7, 95.2	B MCU + Flash	
	C23	100nF	C_0603_1608Metric	99.8, 95.2	B MCU + Flash	
	C24	100nF	C_0603_1608Metric	103.8, 95.2	B MCU + Flash	
	C25	100nF	C_0603_1608Metric	109.2, 77.4	B MCU + Flash	板上印着 R33，尺寸相同会真贴错
	C26	100nF	C_0603_1608Metric	107.8, 95.2	B MCU + Flash	
	C27	100nF	C_0603_1608Metric	111.9, 95.2	B MCU + Flash	
	C28	1uF	C_0603_1608Metric	86.5, 87.7	B MCU + Flash	
	C29	1uF	C_0603_1608Metric	89.5, 87.7	B MCU + Flash	
	C30	100pF	C_0402_1005Metric	120.4, 62.1	E 1090 接收链	板上印着 L14，尺寸相同会真贴错
	C31	100pF C0G	C_0402_1005Metric	106.0, 71.2	E 1090 接收链	
	C32	100pF C0G	C_0402_1005Metric	114.5, 71.2	E 1090 接收链	
	C33	100pF C0G	C_0402_1005Metric	120.9, 71.2	E 1090 接收链	
	C34	100pF C0G	C_0402_1005Metric	129.4, 71.2	E 1090 接收链	
	C35	100pF C0G	C_0402_1005Metric	141.2, 71.2	E 1090 接收链	
	C36	100pF	C_0402_1005Metric	103.0, 66.5	E 1090 接收链	
	C37	100nF	C_0402_1005Metric	109.6, 85.0	E 1090 接收链	
	C38	100pF	C_0402_1005Metric	137.3, 69.2	E 1090 接收链	
	C39	100pF	C_0402_1005Metric	140.0, 83.6	D 978 收发	
	C40	3.6pF	C_0402_1005Metric	123.6, 76.3	D 978 收发	
	C41	2.7pF	C_0402_1005Metric	126.9, 76.3	D 978 收发	
	C42	6.2pF	C_0402_1005Metric	130.2, 76.3	D 978 收发	
	C43	3pF	C_0402_1005Metric	137.6, 78.3	D 978 收发	
	C44	3.6pF	C_0402_1005Metric	121.7, 78.0	D 978 收发	
	C45	100pF	C_0402_1005Metric	119.2, 84.3	D 978 收发	
	C46	3pF	C_0603_1608Metric	118.7, 86.2	E 1090 接收链	板上印着 C47，尺寸相同会真贴错
	C47	200pF	C_0603_1608Metric	118.7, 88.4	E 1090 接收链	板上印着 C46，尺寸相同会真贴错
	C48	1uF	C_0402_1005Metric	99.6, 65.5	E 1090 接收链	

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	C49	100nF	C_0603_1608Metric	55.6, 80.1	E 1090 接收链	
	C51	1nF	C_0603_1608Metric	55.6, 85.2	E 1090 接收链	
	C52	100nF	C_0402_1005Metric	60.4, 96.2	E 1090 接收链	
	C53	100pF	C_0402_1005Metric	110.2, 61.2	E 1090 接收链	
	C54	100pF	C_0402_1005Metric	105.3, 68.1	E 1090 接收链	
	C55	100pF	C_0603_1608Metric	90.2, 74.0	E 1090 接收链	
	C56	100pF	C_0603_1608Metric	85.1, 74.0	E 1090 接收链	
	C57	100pF	C_0402_1005Metric	63.4, 63.3	C 传感器 + GNSS	
	C58	100pF	C_0402_1005Metric	56.5, 64.6	C 传感器 + GNSS	
	C59	100pF	C_0402_1005Metric	56.3, 62.4	C 传感器 + GNSS	
	C60	22uF	C_0805_2012Metric	136.2, 93.1	D 978 收发	
	C61	100nF	C_0603_1608Metric	121.2, 97.0	D 978 收发	
	C62	100nF	C_0603_1608Metric	125.1, 97.0	D 978 收发	
	C63	100nF	C_0603_1608Metric	129.0, 97.0	D 978 收发	
	C64	100nF	C_0603_1608Metric	132.9, 97.0	D 978 收发	
	C65	100nF	C_0603_1608Metric	136.8, 97.0	D 978 收发	
	C66	22uF	C_0805_2012Metric	123.2, 100.5	D 978 收发	
	C67	1uF	C_0603_1608Metric	126.9, 93.1	D 978 收发	
	C68	18pF	C_0603_1608Metric	123.1, 93.8	D 978 收发	
	C69	18pF	C_0603_1608Metric	127.7, 100.5	D 978 收发	
	C81	470pF C0G	C_0402_1005Metric	118.6, 73.5	E 1090 接收链	
	C82	100nF	C_0402_1005Metric	83.0, 84.5	B MCU + Flash	
	C83	100nF	C_0402_1005Metric	93.6, 79.5	B MCU + Flash	
	C84	1uF	C_0402_1005Metric	93.6, 82.0	B MCU + Flash	
	C85	100nF	C_0402_1005Metric	96.0, 79.5	B MCU + Flash	
	C86	100nF	C_0402_1005Metric	96.0, 82.0	B MCU + Flash	
	C87	10nF C0G	C_0603_1608Metric	97.8, 84.4	B MCU + Flash	
	R1	10k	R_0603_1608Metric	79.0, 87.2	C 传感器 + GNSS	
	R2	4.7k	R_0603_1608Metric	61.4, 106.1	C 传感器 + GNSS	
	R3	4.7k	R_0603_1608Metric	61.4, 108.7	C 传感器 + GNSS	
	R4	1k	R_0603_1608Metric	110.6, 87.7	B MCU + Flash	
	R5	10k	R_0603_1608Metric	95.7, 98.3	B MCU + Flash	
	R6	10k	R_0603_1608Metric	99.8, 98.3	B MCU + Flash	
	R7	5.1k	R_0603_1608Metric	103.8, 98.3	B MCU + Flash	
	R8	5.1k	R_0603_1608Metric	107.8, 98.3	B MCU + Flash	
	R9	27R	R_0603_1608Metric	92.5, 87.7	B MCU + Flash	
	R10	27R	R_0603_1608Metric	95.6, 87.7	B MCU + Flash	
	R11	3.32k	R_0402_1005Metric	98.8, 68.0	E 1090 接收链	
	R17	10k	R_0603_1608Metric	121.4, 105.2	D 978 收发	
	R18	10k	R_0603_1608Metric	136.2, 62.1	E 1090 接收链	
	R19	52.3R	R_0402_1005Metric	138.5, 71.2	E 1090 接收链	
	R21	0R	R_0603_1608Metric	115.2, 85.0	E 1090 接收链	
	R22	0R	R_0603_1608Metric	87.5, 71.2	E 1090 接收链	
	R23	0R	R_0603_1608Metric	91.4, 71.2	E 1090 接收链	
	R24	0R DNP(旁路外接)	R_0402_1005Metric	112.7, 66.0	不贴 (DNP)	
	R25	0R DNP(旁路板载)	R_0402_1005Metric	114.8, 66.0	不贴 (DNP)	
	R26	10k	R_0603_1608Metric	58.8, 76.4	C 传感器 + GNSS	
	R27	10k	R_0603_1608Metric	61.8, 73.3	C 传感器 + GNSS	
	R30	1k DNP	R_0402_1005Metric	108.0, 66.6	不贴 (DNP)	

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	R31	1k	R_0603_1608Metric	115.0, 87.3	E 1090 接收链	
	R32	10k	R_0603_1608Metric	112.5, 75.2	E 1090 接收链	
	R33	100k	R_0603_1608Metric	108.7, 75.2	E 1090 接收链	板上印着 C25, 尺寸相同会真贴错
	R34	10k	R_0603_1608Metric	55.6, 77.5	E 1090 接收链	
	R35	10k	R_0603_1608Metric	55.6, 82.6	E 1090 接收链	
	R36	1k DNP	R_0603_1608Metric	61.4, 103.5	不贴 (DNP)	
	R47	10k	R_0603_1608Metric	132.5, 85.0	D 978 收发	
	R50	10k	R_0603_1608Metric	94.0, 85.2	B MCU + Flash	
	R51	10k	R_0603_1608Metric	96.0, 85.2	B MCU + Flash	
	R52	10k	R_0402_1005Metric	60.0, 83.0	C 传感器 + GNSS	
	R53	10k	R_0402_1005Metric	63.0, 84.4	C 传感器 + GNSS	
	R54	10R	R_0402_1005Metric	109.6, 82.1	E 1090 接收链	
	R55	10k	R_0603_1608Metric	59.1, 80.3	C 传感器 + GNSS	
	R56	10k	R_0603_1608Metric	132.3, 89.7	D 978 收发	
	R57	33R	R_0603_1608Metric	85.7, 95.3	E 1090 接收链	
	L1	18nH 0402CS-18NXGRW	L_0402_1005Metric	105.5, 69.8	E 1090 接收链	
	L2	33nH	L_0402_1005Metric	61.8, 64.8	C 传感器 + GNSS	
	L7	6.8uH	L_0805_2012Metric	131.5, 93.1	D 978 收发	
	L8	100nH	L_0402_1005Metric	142.6, 87.0	D 978 收发	
	L9	7.5nH	L_0402_1005Metric	124.2, 78.0	D 978 收发	
	L10	27nH	L_0402_1005Metric	120.8, 76.3	D 978 收发	
	L11	6.8nH	L_0402_1005Metric	134.2, 76.3	D 978 收发	
	L12	6.8nH	L_0402_1005Metric	137.6, 76.3	D 978 收发	
	L13	7.5nH	L_0402_1005Metric	119.2, 83.2	D 978 收发	
	L14	100nH	L_0402_1005Metric	120.4, 63.3	E 1090 接收链	板上印着 C30, 尺寸相同会真贴错
	L15	33nH	L_0402_1005Metric	58.0, 67.6	C 传感器 + GNSS	
	D1	B5819W	D_SOD-123	97.8, 91.3	B MCU + Flash	有极性, 认阴极标记带
	D2	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	128.3, 62.1	E 1090 接收链	
	D3	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	140.0, 86.9	D 978 收发	
	D4	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	60.0, 90.4	B MCU + Flash	
	D5	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	60.0, 93.6	B MCU + Flash	
	F2	6V/200mA	Fuse_0805_2012Metric	129.8, 105.2	D 978 收发	
	F3	6V/200mA	Fuse_0805_2012Metric	140.2, 62.1	E 1090 接收链	
	F4	6V/200mA	Fuse_0805_2012Metric	61.8, 67.4	C 传感器 + GNSS	
	F5	6V/200mA	Fuse_0805_2012Metric	56.4, 69.7	C 传感器 + GNSS	
	Q2	AO3401A	SOT-23	125.1, 105.2	D 978 收发	
	Q3	AO3401A	SOT-23	132.0, 62.1	E 1090 接收链	
	Q4	AO3401A	SOT-23	61.8, 70.2	C 传感器 + GNSS	
	Q5	AO3401A	SOT-23	57.8, 72.6	C 传感器 + GNSS	
	U1	ME6211C33M5	SOT-23-5	69.4, 93.3	A 电源	认 pin1 (缺角/圆点)
	U2	TPS7A2033PDBVR	SOT-23-5	63.1, 93.3	A 电源	认 pin1 (缺角/圆点)
	U3	TPS7A2033PDBVR	SOT-23-5	74.3, 93.3	A 电源	认 pin1 (缺角/圆点)
	U4	BNO085	BNO085_LGA-28	63.7, 78.3	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U5	BMP388	BMP388_LGA-10	71.3, 81.2	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U6	QMC5883P	LGA-16_3x3mm_P0.5mm	71.3, 77.0	C 传感器 + GNSS	认 pin1 (缺角/圆点)

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	U7	ATGM336H-6N-74	ATGM336H_LCC-18	70.4, 66.4	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	U8	RP2040	QFN-56-1EP_7x7mm_P0.4mm_EP3.2	88.4, 80.5	B MCU + Flash	认 pin1 (缺角/圆点)
	U9	W25Q128JVS IQ	SOIC-8_5.3x5.3mm_P1.27mm	103.4, 82.1	B MCU + Flash	认 pin1 (缺角/圆点)
	U10	CC1312R1F3RGZR	QFN-48-1EP_7x7mm_P0.5mm_EP5.1	126.2, 85.0	D 978 收发	认 pin1 (缺角/圆点)
	U11	QPL9547TR7	DFN-8-1EP_2x2mm_P0.5mm_EP0.8	101.6, 71.2	E 1090 接收链	认 pin1 (缺角/圆点)
	U12	BGA2817	SOT-363_SC-70-6	117.7, 71.2	E 1090 接收链	认 pin1 (缺角/圆点)
	U14	AD8313ARMZ	MSOP-8_3x3mm_P0.65mm	114.1, 82.1	E 1090 接收链	认 pin1 (缺角/圆点)
	U15	TLV3501(TOKMAS)	SOT-23-6	83.1, 91.3	E 1090 接收链	认 pin1 (缺角/圆点)
	U16	XA17-G4K	SOT-363_SC-70-6	114.6, 62.1	E 1090 接收链	认 pin1 (缺角/圆点)
	U17	XA17-G4K	SOT-363_SC-70-6	58.8, 63.3	C 传感器 + GNSS	认 pin1 (缺角/圆点)
	FL1	TA0970A	TA0970A_SMD3838-6	110.2, 71.2	E 1090 接收链	SAW 滤波器分输入输出, 贴反不工作
	FL2	TA0970A	TA0970A_SMD3838-6	125.1, 71.2	E 1090 接收链	SAW 滤波器分输入输出, 贴反不工作
	Y1	12MHz CL=10pF ABM8-272-T3	Crystal_SMD_3225-4Pin_3.2x2.5mm	100.4, 87.7	B MCU + Flash	无源晶体, 转 180° 等价, 长边对长边即可
	Y2	48MHz ABM8W-7pF	Crystal_SMD_3225-4Pin_3.2x2.5mm	133.8, 81.7	D 978 收发	无源晶体, 转 180° 等价, 长边对长边即可
	Y3	32.768kHz FC-135	Crystal_SMD_3215-2Pin_3.2x1.5mm	122.7, 91.3	D 978 收发	
	J1	J3_HAT_2x20_SMD排针	PinHeader_2x20_P2.54mm	92.6, 106.3	F 对外接口	
	J2	U.FL_GNSS_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	52.9, 74.1	C 传感器 + GNSS	
	J4	USB-C_16P	USB_C_Receptacle_HRO_T C-31-M-12	53.5, 92.0	B MCU + Flash	
	J5	U.FL_978	U.FL_Hirose_U.FL-R-SMT-1_Vertical	145.5, 83.5	D 978 收发	
	J6	U.FL_1090_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	124.3, 62.1	E 1090 接收链	
	J7	U.FL_IFA_TEST(π后调试口)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	103.1, 61.3	E 1090 接收链	
	J8	U.FL→内置patch	U.FL_Hirose_U.FL-R-SMT-1_Vertical	52.9, 64.4	C 传感器 + GNSS	
	ANT1	IFA_1090	ANT_IFA_1090MHz	85.5, 57.8	不贴 (板载天线)	
	ZP1	DNP 并-天线侧	C_0603_1608Metric	85.5, 63.6	不贴 (DNP)	
	ZP2	DNP 并-电台侧	C_0603_1608Metric	90.5, 63.6	不贴 (DNP)	
	ZS1	OR 串	L_0603_1608Metric	88.0, 62.9	E 1090 接收链	
	SW1	RESET短接焊盘	SolderJumper-2_P1.3mm_Open_Pad1.0x1.5	88.7, 91.3	不贴 (短接焊盘)	
	SW2	BOOTSEL短接焊盘	SolderJumper-2_P1.3mm_Open_Pad1.0x1.5	92.9, 91.3	不贴 (短接焊盘)	
	TP1	TestPoint	TestPoint_Pad_1.0x1.0mm	102.0, 91.3	不贴 (测试点焊盘)	
	TP2	TestPoint	TestPoint_Pad_1.0x1.0mm	104.9, 91.3	不贴 (测试点焊盘)	
	TP3	TestPoint	TestPoint_Pad_1.0x1.0mm	107.8, 91.3	不贴 (测试点焊盘)	
	TP4	TestPoint	TestPoint_Pad_1.0x1.0mm	110.7, 91.3	不贴 (测试点焊盘)	
	TP5	TestPoint	TestPoint_Pad_1.0x1.0mm	113.5, 91.3	不贴 (测试点焊盘)	
	TP6	TestPoint	TestPoint_Pad_1.0x1.0mm	116.4, 91.3	不贴 (测试点焊盘)	
	TP7	TestPoint	TestPoint_Pad_1.0x1.0mm	80.7, 78.7	不贴 (测试点焊盘)	
	H1	—	MountingHole_2.7mm_M2.5	54.0, 56.0	不贴 (安装孔)	
	H2	—	MountingHole_2.7mm_M2.5	146.0, 56.0	不贴 (安装孔)	
	H3	—	MountingHole_2.7mm_M2.5	54.0, 106.0	不贴 (安装孔)	
	H4	—	MountingHole_2.7mm_M2.5	146.0, 106.0	不贴 (安装孔)	